

Read $f(2)$ and the preimages of 2.5.

Ans. $f(2) = 0$; preimages of 2.5: -3 and 3.

Expression used for the plot: f $(x) = 0{,}5x^2 - 2$.

5. Running thread · Method [ALG] Procedure: choose a number; multiply it by -2; add 10; divide by 2. Result for 7? Expression in x ?

Ans. $\dfrac{-14+10}{2} = -2$; in general $\dfrac{-2x+10}{2} = -x+5$.

Sheet 74 Mon. 31 May 2027 False positives and false negatives

1. Warm-up · Flash [PRB] Define "false positive" and "false negative".

Ans. False positive: positive test for a person who is not ill. False negative: negative test for a person who's ill.

2. Core · Method [PRB] 10,000 people, 2% ill; the test is positive for 90% of the ill and 10% of those without the disease. How many false negatives, false positives and positives in total?

Ans. Ill 200: 180 positive, so 20 false negatives; not ill 9,800: 980 false positives; positives: $180 + 980 = 1\,160$.

3. Core + · Interpret [PRB] With the previous data, calculate the proportion of people truly ill among the positive ones and comment.

Ans. $\dfrac{180}{1\,160} \approx 15.5\%$: since the disease is rare, most positives are false positives.

4. Back · True/false [PNP] True or false? Justify. The area of a square is proportional to the length of its side.

Ans. False. $c = 1 \mapsto 1$ and $c = 2 \mapsto 4 \neq 2 \times 1$.

5. Running thread · Refute [LIT] Conjecture: "for every integer n, $n^2 + n + 41$ is prime". Test $n = 40$.

Ans. $40^2 + 40 + 41 = 1681 = 41^2$: not prime. The conjecture is false.

Successful tests prove nothing; a failed test refutes.

Consolidation, fluency and transition to Première

Sheet 75 Tue. 1 st June 2027 Synthesis problems drawing on several areas

1. Warm-up · Flash [EQ] Solve $3(x-1) = x+5$.

Ans. $3x - 3 = x + 5$, $2x = 8$, $x = 4$.

2. Core · Method [GEO] For which value of k are points $A(1;1)$, $B(3;5)$ and $C(k;9)$ collinear?

Ans. $\overrightarrow{AB}\begin{pmatrix}2\\4\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}k-1\\8\end{pmatrix}$: $2 \times 8 - 4(k-1) = 0$, therefore $k = 5$.

3. Core + · Method [PRO] A price rises by 10% then falls by 10%. What is the overall change?

Ans. $1{,}1 \times 0{,}9 = 0.99$: a decrease of 1%.

4. Back · Flash [PER] Total surface area of a cuboid with dimensions 4, 3, 2 (cm)?

Ans. $2(4 \times 3 + 4 \times 2 + 3 \times 2) = 52$ cm^2.

5. Running thread · Flash [SOL] Calculate the exact volume of a sphere with radius 3 cm.

Ans. $V = \dfrac{4}{3}\pi \times 27 = 36\pi$ cm^3.

Sheet 76 Thu. 3 June 2027 Review of the six skills

1. Warm-up · Metacognition [MET] State the six mathematical skills developed in lycée.

Ans. Investigate, model, represent, reason, calculate, communicate.

2. Core · Metacognition [MET] Match: "express a statement as an equation"; "draw a curve"; "justify an equality by a written calculation" to a skill.

Ans. Model; represent; reason (or communicate, depending on the emphasis on written work).

3. Core + · Metacognition [MET] To solve 2 $x+3=11$, which skill is mainly used?

Ans. Calculate (rearrange a linear equation).

To be discussed orally: setting up an equation for a problem falls under "modelling".

4. Back · Problem [MOD] Water at 20 °C heats up by 3 °C per minute: $T(t) = 20+3t$. After how many minutes does it take to reach 65 °C?

Ans. $20+3t=65 \Leftrightarrow t=15$ min.

5. Running thread · Flash [EQ] Solve in $\mathbb{R}$: $0{,}5x+1{,}2=2{,}7$.

Ans. $x=3$

Sheet 77 Mon. 7 June 2027 Targeted revisiting of prerequisites for Première

1. Warm-up · Flash [DEV] Expand $(x+3)^2$.

Ans. x^2+6x+9.

2. Core · Method [EQ] Check that $x^2-5x+6=(x-2)(x-3)$ and deduce the solutions of $x^2-5x+6=0$.

Ans. $(x-2)(x-3)=x^2-5x+6$; zero product: $x=2$ or $x=3$.

3. Core + · Anticipation [MOD] For $f(x)=x^2$, calculate the rate of change $\dfrac{f(3)-f(1)}{3-1}$.

Ans. $\dfrac{9-1}{}=4$: this prepares for the derivative at a point in Première.

A Première-level idea; a simple calculation reusing earlier skills.

4. Back · Construction steps [CON] We place M and N midpts of $[AB]$ and $[AC]$ then we draw $[MN]$. What can be said about (MN) and (BC) ? Calculate MN if $BC=9$ cm.

Ans. $(MN)\parallel(BC)$ and $MN=\dfrac{9}{2}=4.5$ cm.

5. Running thread · Write up [DEM] Prove that a triangle with sides 7, 24 and 25 is right-angled.

Ans. $7^2+24^2=49+576=625=25^2$: converse of Pythagoras' theorem.

Sheet 78 Tue. 8 June 2027 Synthesis problems: choosing and justifying a strategy

1. Warm-up · Code reading [ALG] What does $f(2)$?

```
def f(x):
    return x**2 - 3*x + 1
```

Ans. $4-6+1=-1$.

2. Core · Completion [ALG] Complete so that the function returns True when A, B, C are collinear:

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == ....
```

Ans. The determinant is compared to 0: `== 0`.

3. Core + · Completion [ALG] Complete the absolute value function:

```
def vabs(x):
    if x >= 0:
        return x
    else:
        return ....
```

Ans. return -x .